Yeast Fermentation

Student worksheet | Balloon lab | Biology 2e, Chapter 7

1. Plan before you mix

Names: _____ Class/date: _____

Pair/team: _____ Pooled set ID: _____ Assigned flask IDs: _____

Question: How does the available sugar affect gas accumulation in a yeast culture? Prepare your assigned one or two flasks, then pool three no-added-sugar, three glucose and three sucrose trials.

A balloon collects gas. Measure the distance around its widest part. Circumference is a relative indicator of gas accumulation, not a volume in milliliters.

Teacher-selected setup

Setting	Your assignment
Method / measurement / unit	Balloon lab / Balloon circumference / cm
Flask capacity / final liquid	____ mL / ____ mL
Treatment stock / yeast suspension per flask	____ mL / ____ mL
Sugar stock / yeast stock concentration	_____ / _____
Temperature / interval / end time	____ C / ____ min / ____ min

Prediction: Compare the three treatments and explain your reasoning.

What will you change and measure? Name three things to keep the same.

Revised classroom target: 20 g yeast + 80 g sugar per 1,000 mL total water. Stock options are 20 g yeast / 500 mL water and 80 g sugar / 500 mL water; or 40 g and 160 g, respectively, per 1,000 mL water each. Water-added amounts are not exact final solution volumes.

Confirm IDs, stock labels, volumes and timing with your teacher. Wear eye protection, do not taste cultures, and never seal a culture in a rigid closed container. Follow the teacher's equipment setup instructions.

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2. Prepare and record your own flasks

Label each flask and add its assigned treatment stock. Have the yeast stock gently remixed before taking your aliquot. Add yeast last, mix consistently, promptly connect the balloon, and start that flask's timer. Use a fresh mixture per flask.

Gently measure the widest circumference without squeezing. Use the same measurement method each time. Record NI (not inflated) for a limp balloon; do not invent a zero.

NI = not inflated; M = missing; L = suspected leak. Do not replace NI or M with zero.

Flask A ID: ____ Treatment: _____ Start time: ____ Temp: ____ C

Flask B ID: ____ Treatment: _____ Start time: ____ Temp: ____ C

Measurement: Balloon circumference (cm). Record actual circumference in each column.

Elapsed min*	A (cm)	B (cm)	Notes
0			
10			
20			
30			
40			
50			
60			

*Use the teacher-assigned schedule if different. Each flask has its own elapsed time. Leave B unused if you have only one flask. Record first visible inflation separately: A ____ min; B ____ min.

Describe problems or observations; keep raw readings and explain any flags.

Never use a rigid cap. The teacher stops setups if foam enters a balloon or its inflation becomes excessive.

Share flask IDs, readings, units, temperature and notes. Nine separate flasks are nine trials; repeated time readings from one flask are still one trial.

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3. Pool three trials of each treatment

Set ID: _____ Measurement: circumference (cm)

Pool matching recipes, flask sizes and equipment. Keep virtual data separate. NI = not inflated; M = missing; L = suspected leak. Do not replace NI or M with zero.

No added sugar

Elapsed min	N1	N2	N3	Mean (cm)	n / NI count
0					
10					
20					
30					
40					
50					
60					

Glucose

Elapsed min	G1	G2	G3	Mean (cm)	n / NI count
0					
10					
20					
30					
40					
50					
60					

Sucrose

Elapsed min	S1	S2	S3	Mean (cm)	n / NI count
0					
10					
20					
30					
40					
50					
60					

Mean = sum of comparable numeric readings / number of readings (n). Report NI separately; a numeric-only mean with n below 3 does not represent all three flasks. Retain flagged readings and explain exclusions. Relabel times if the teacher assigns a different schedule.

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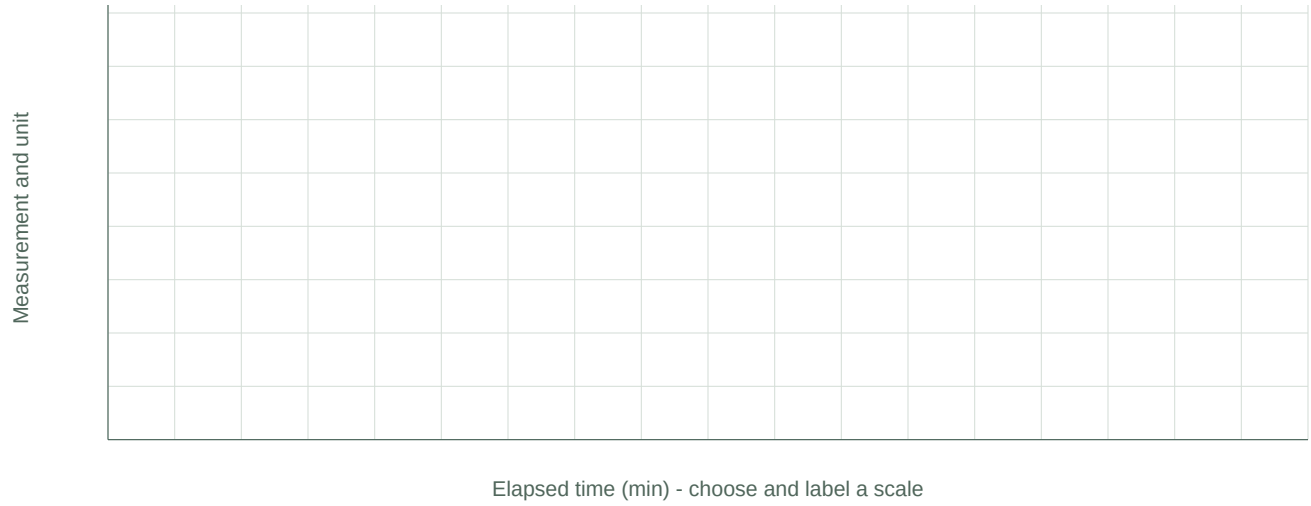
4. Compare evidence and explain

Graph circumference (cm) versus elapsed minutes. Label scales and units. Use different colors or symbols and a legend for the treatments. Mark missing/NI data as gaps, not zero; label means with n when below three.



Legend: No sugar _____ Glucose _____ Sucrose _____

1. Compare treatments using numbers, units and variation among trials. Does the treatment with the highest final reading also have the fastest increase over the final interval?

2. Why include yeast with no added sugar? What would a sugar-without-yeast control help test?

3. Identify one limitation of this collection method. Why can these readings alone not measure ATP yield or prove oxygen was absent?

After the comparison: propose a follow-up variable and controls. Obtain teacher approval; keep new results separate.